

6.2 Changing population structures

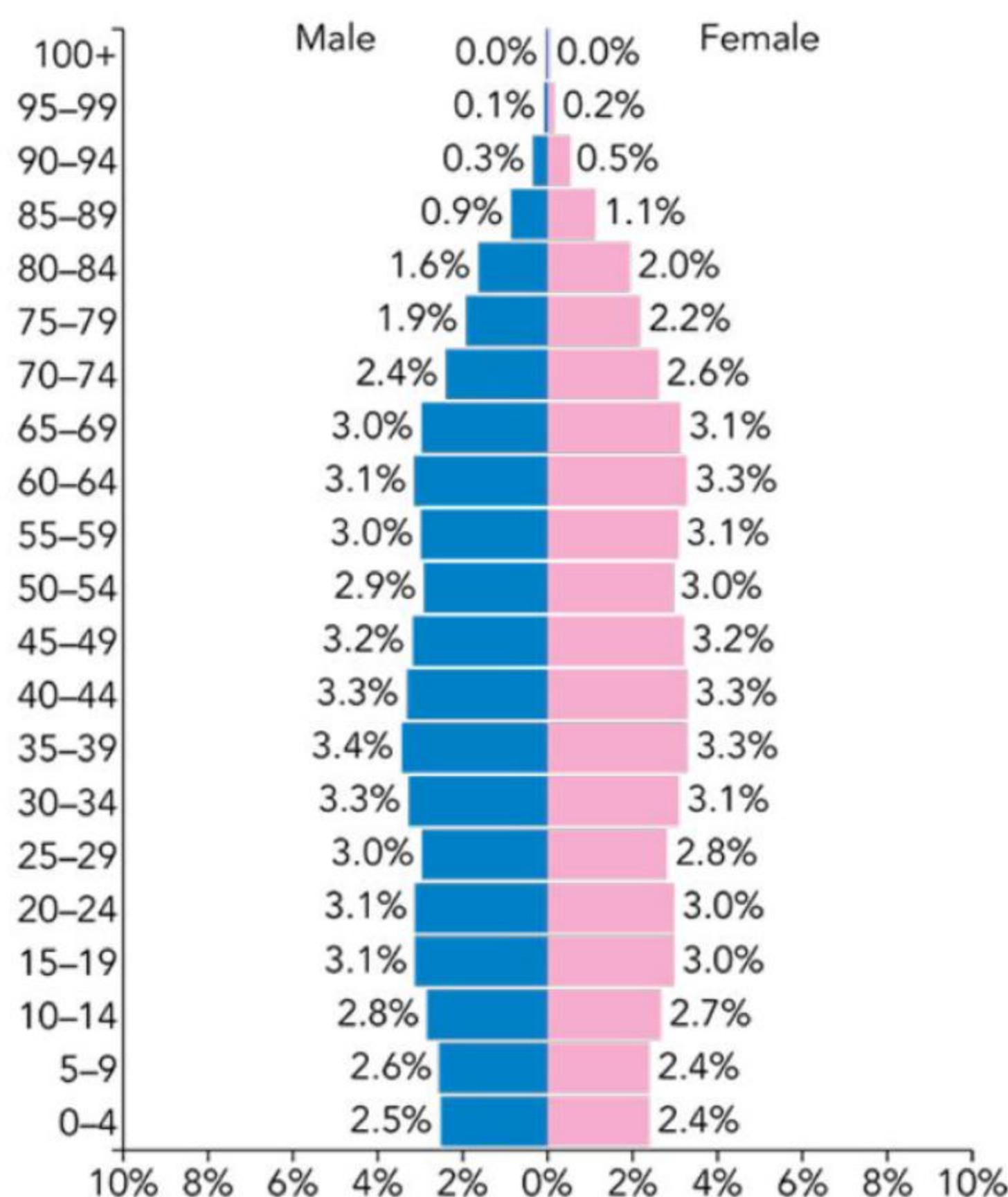
Population structure refers to the division of a population into a series of groups according to age and sex. Population structures in LICs and HICs are very different. **Population pyramids** can help us to understand the population structure for a country. The shape of a population pyramid helps us interpret the birth rate, death rate, and life expectancy for a country.

The pyramids divide people into cohorts (age groups) and then place them on either side of the vertical axis, with males on one side and females on the other. They show the size of each age group as an actual number (in thousands of millions) or as a percentage of the total population. There are three main age groups in a population pyramid:

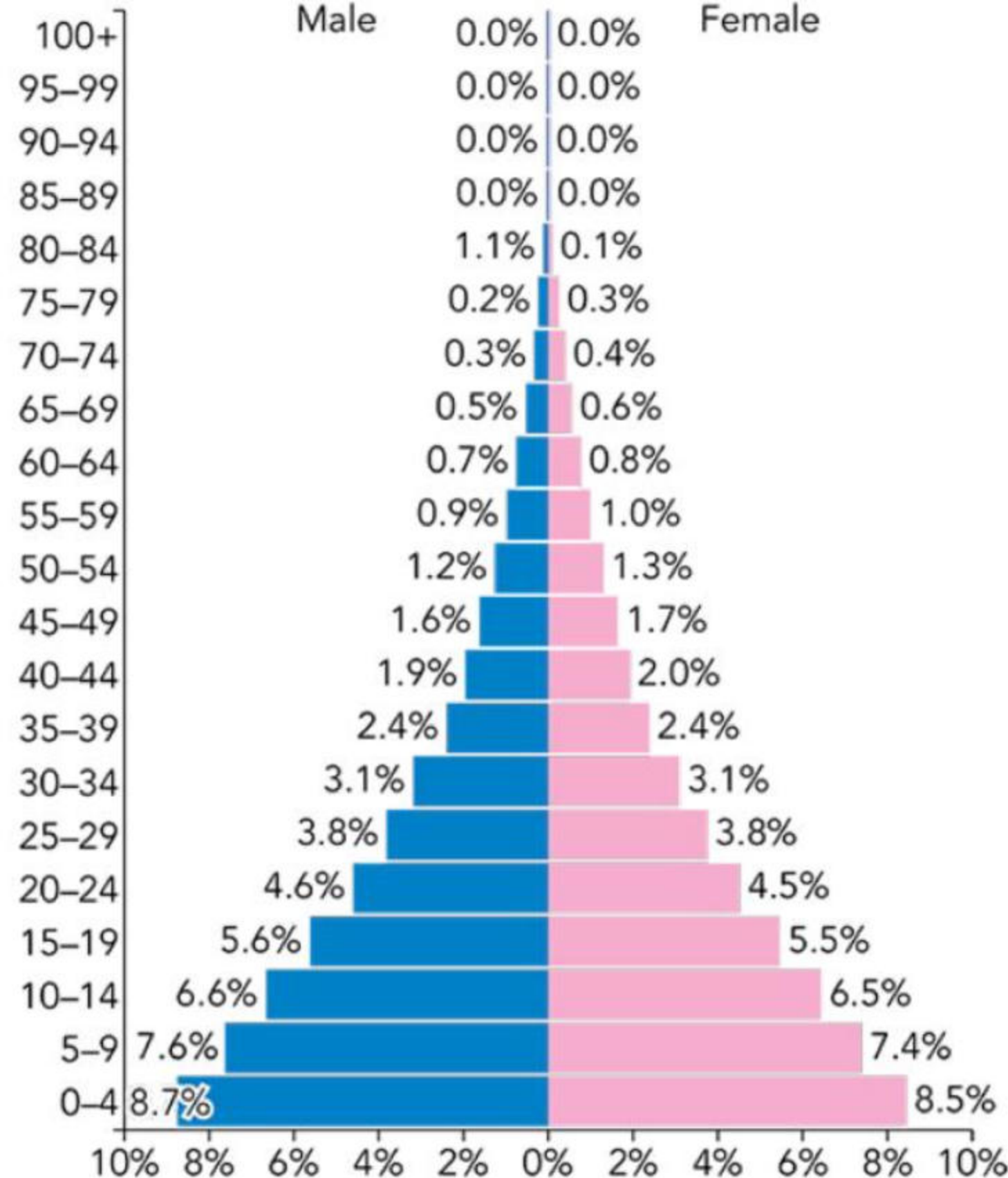
- **Young dependents:** people aged 0–14, who depend on others for resources.
- **Economically active:** people aged 15–64, who are usually working and providing resources to young and elderly dependents.
- **Elderly dependents:** people aged 65 upwards, who are not usually in work.

Population structures are dynamic and can indicate future population trends. The main factors influencing population structures are natural increase and net migration. Natural increase will first increase the number of people aged 0–14 (shown by a wide and expanding base on a pyramid showing a country's population structure). The effects of wars, natural disasters, and epidemics on particular age groups will be shown by 'indentations' on a pyramid showing a country's population structure. Countries that have positive net migration will show a 'bulge' in the proportion of economically active people, especially between the ages of 20–40.

When comparing the pyramids of different countries, important characteristics can offer clues about their relative levels of economic development. A wide base for a country indicates a relatively high birth rate but low life expectancy. This is typical of an LIC (Source B). Concave (inward curving) sides to a country's population pyramid can indicate low life expectancy and a high death rate. Perpendicular (at right angles) sides relative to the base of a country's pyramid tell us that the majority of those born survive to an old age. This is typical of an HIC (Source A).



Source **A**: Population pyramid for UK, projected 2030



Source B: Population pyramid for Somalia, projected 2030

TASK 1: Study Sources **A** **B**

Source A shows the population pyramid for the UK, a typical HIC. Source B shows that of Somalia, a typical LIC.

- Estimate the percentage** of children 0–14
 - in the UK and
 - in Somalia.
- Estimate the percentage** of elderly over the age of 65
 - in the UK and
 - in Somalia.
- What does the difference between these numbers in the two countries show?
- Describe three differences in the shape and structure** of the two population pyramids.

Ageing and youthful population structures

A population pyramid can therefore tell us whether a country has an **ageing population** structure or a **youthful population** structure.

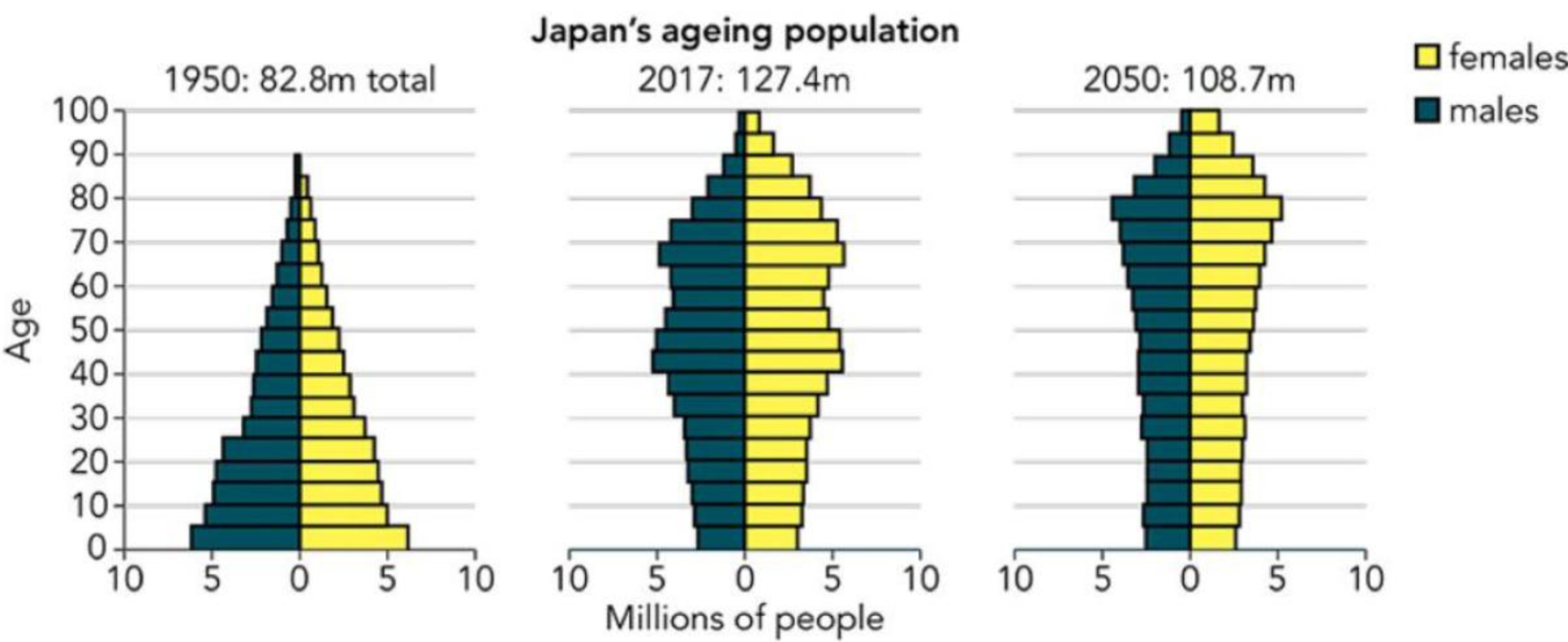
Ageing population structure

In some countries, the birth rate has fallen below the death rate. This is called an ageing population. Countries such as Germany and Japan are in Stage 5 of the DTM. During the 20th century, the world’s HICs experienced significant changes in average family size and the proportion of people surviving to an older age, giving rise to this phenomenon. In Japan, Iceland, Italy, Australia, Germany, and many other countries, life expectancy is now 80 years or higher. More than 20% of the population in these countries are currently aged 65 or over. In the future, the proportion of older citizens in HICs will grow even higher.

An ageing population has specific social, economic, and environmental impacts for a country (Table 6.3). Planning for the ‘demographic time bomb’ (where population decline means there are not enough young people to replace the older population as they die out) is one of the greatest political challenges of our times.

Table 6.3: Impacts of an ageing population

Positive impacts of an ageing population	Negative impacts of an ageing population
Many older people continue to work into their 70s and 80s. Older workers have plenty of experience and insights that young people may lack.	High government public spending on the elderly diverted from other projects. For example, in Japan, 40% of government spending is on the elderly.
Older people have invaluable family, community, and civil society roles to play. They may help with care for grandchildren and/or work in volunteering.	The working population is shrinking fast which has a negative impact on economic growth.
Some industries including leisure and tourism may benefit from elderly spending.	A rise in home care by families who may have to give up full-time jobs to become carers.
Older people tend to consume less resources and therefore reduce environmental pressures on land, water, and energy.	Greater pension costs meaning a greater tax burden on the working population.
An ageing population could increase policy demand for designing accessible, safe, and green public spaces, promoting physical activity and improving physical and mental health.	Older people will be more vulnerable to climate-related environmental stresses, such as heatwaves and floods, putting more pressure on care and emergency services.



Source C: Population pyramids showing Japan’s ageing population

TASK 2: Study Source C

a **Estimate the predicted number** of elderly dependents in Japan in 2050.

b Use the three population pyramids to **describe the changes** in Japan’s ageing population over time

c Japan is a HIC. Suggest why death rates are low but birth rates are lower.



Source D: Elderly people in the UK



Source E: Elderly people in the UK

TASK 3: Study Sources D E

Discuss the following questions in pairs:

- a** Identify one economic challenge and one social challenge for a country with an ageing population.
- b** Explain why they are a challenge for the government of a country.
- c** Study Sources D and E. Give two reasons why an ageing population can bring benefits to a country.
- d** What strategies can a government use to try to cut the economic costs of an ageing population?

Youthful population structure

Many LICs have high birth rates (Stage 3 of the DTM), resulting in a high proportion of under 15s in the population. This is called a youthful population. Youthful populations have very high birth rates, slowly decreasing death rates, and slow rises in life expectancy. As with an ageing population, a youthful population has specific social, economic, and environmental impacts for a country.

A youthful population can provide a large and cheap future workforce (if employment opportunities are available). This workforce can provide a growing market for manufactured products and a large tax base for the country. However, it can put strains on education and health services, with too many youth dependents compared to adults, like teachers and doctors, who can support them. This can become an economic problem that will put a strain on food and water supplies for both families and the country. Malnutrition may be an issue, and due to poverty, more people turn to using cheap forest wood for heating, cooking, and building, contributing to desertification.

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TASK 4

- a** Create a table listing the economic, social, and environmental challenges and opportunities of a country with a youthful population. You can carry out research to supplement this table.
- b** Uganda has the youngest population in the world. Carry out research to answer the following questions:
 - i** What percentage of the Ugandan population are below the age of 30?
 - ii** Why does Uganda have such a youthful population?
 - iii** What are some of the problems and some of the benefits of Uganda's youthful population?

REFLECTION

What strategy do you use when you are asked to find the impacts of something in geography? For example, do you think of all of the impacts and then classify each one as an advantage or a disadvantage? Or do you categorise the impacts into groups such as economic, social, political, and environmental? Are there any other ways?

Next time you do a task about the impacts of something in geography, look to see what strategy you are using. Try using another strategy and see what works best for you.

PROJECT

Country A's population is ageing. By 2040, nearly one in seven people is projected to be aged over 75. This will have a major effect on Country A. Below

are some viewpoints on this issue:

‘Ageing will increase the amount of money we need to spend on health and well-being but it will not be enough. We need to spend more on health, ill-health, and disability. We need more hospitals, medical care, doctors and nurses, and care homes. Pensions and other costs will also increase. Where will the money come from?’ – Minister for Health.

‘Enabling people to work for longer will help society support the growing number of dependents and provide people with the financial resources for increasingly long retirements. Lifelong learning brings benefits to individuals, employers, and wider society that will be increasingly valuable in an ageing population. We need to raise the retirement age to 75, but who wants that?’ – Minister for Work and Pensions.

‘There are more people over 65 than under 16 in the country. If it wasn’t for immigrants, we would have a real problem filling the jobs we need doing. We must relax our immigration laws to boost our economy but will this be controversial?’ – Interior Minister.

Imagine you are part of a team in charge of advising Country A’s government about how best to respond to its ageing population. Write a one-page proposal outlining your main suggestions.